



Embedded

FA 2026 • 2026-10-05

Embedded 103: Breadboarding and Hardware

Swetha and Tony

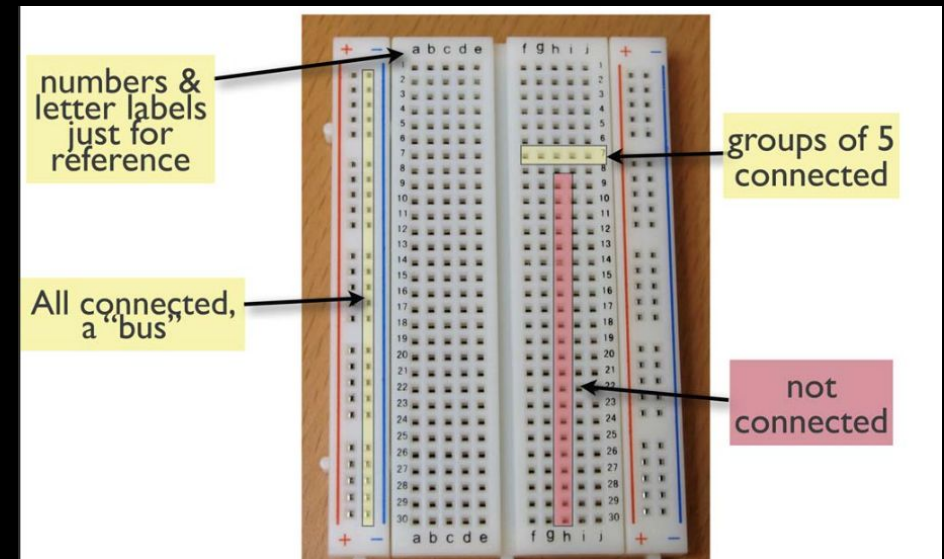
Announcements

- CSAW Embedded Security Challenge
 - Using AI and LLMs for hardware security
 - We have two teams that qualified for the final round!
 - Team 1: Minh, Shovan, Krishnan, Cygnus
 - Team 2: Jake, Swetha, Nikhil, Ryan



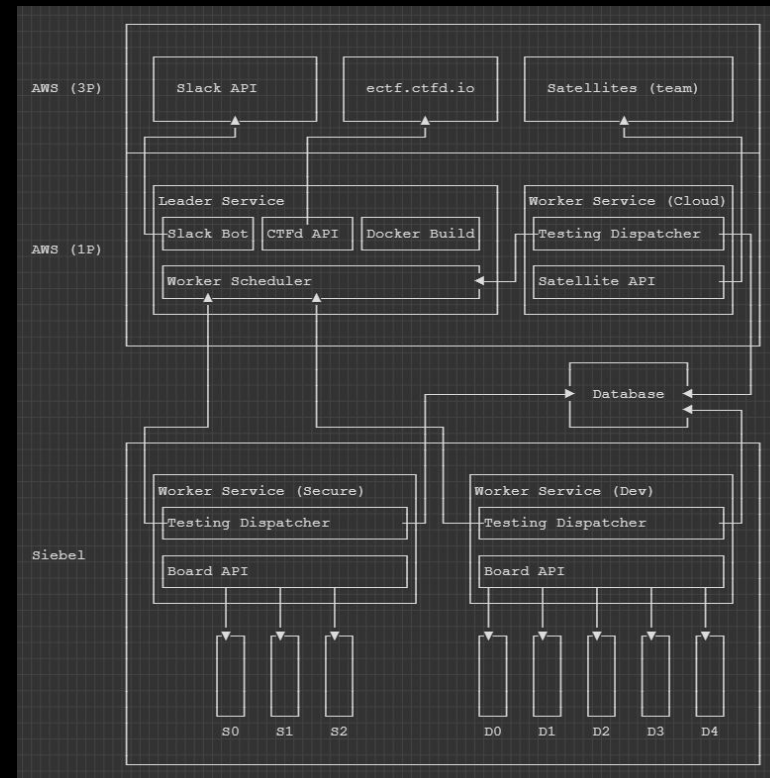
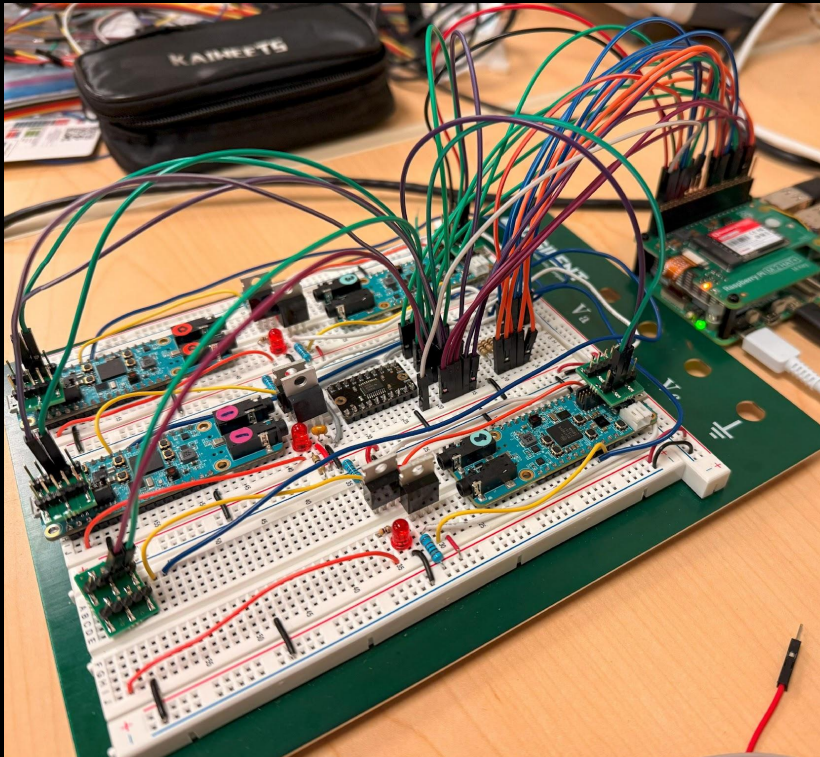
What is breadboarding?

- Process of building and testing circuits on a breadboard (as opposed to soldering)
- Allows us to observe circuit behavior and debug in real time
- Circuits used to be prototyped on wooden boards used to cut bread
- **Breadboard** - plastic board with metal sockets
 - Power rail, ground rail, horizontal rails



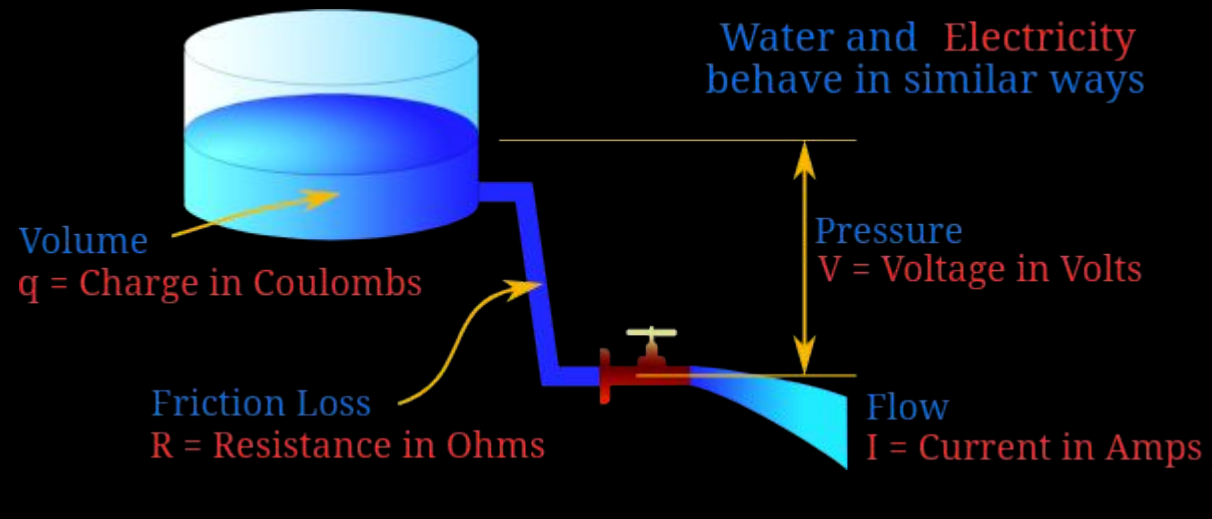
How is breadboarding relevant?

- Understanding how circuits work is important for understanding voltage glitching and fault injection
- Helps with quick prototyping, debugging, instrumenting boards



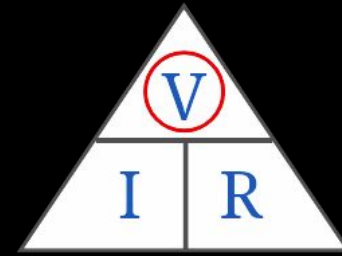
What is electricity?

- Current (amperage)
 - Electrons over time
 - Think: Flow of water through a pipe
- Voltage
 - Potential energy per electron (kinda)
 - Flows from high to low, like potential energy due to gravity
 - Think: Pressure of water in a pipe
- Resistance
 - Voltage drop per amp
 - Think: "loss" pressure due to long or narrow pipes

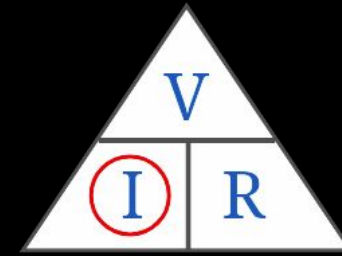


Ohm's Law

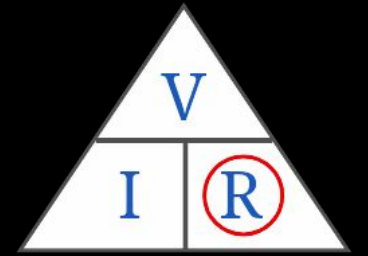
Resistance is a proportional voltage drop proportional to current



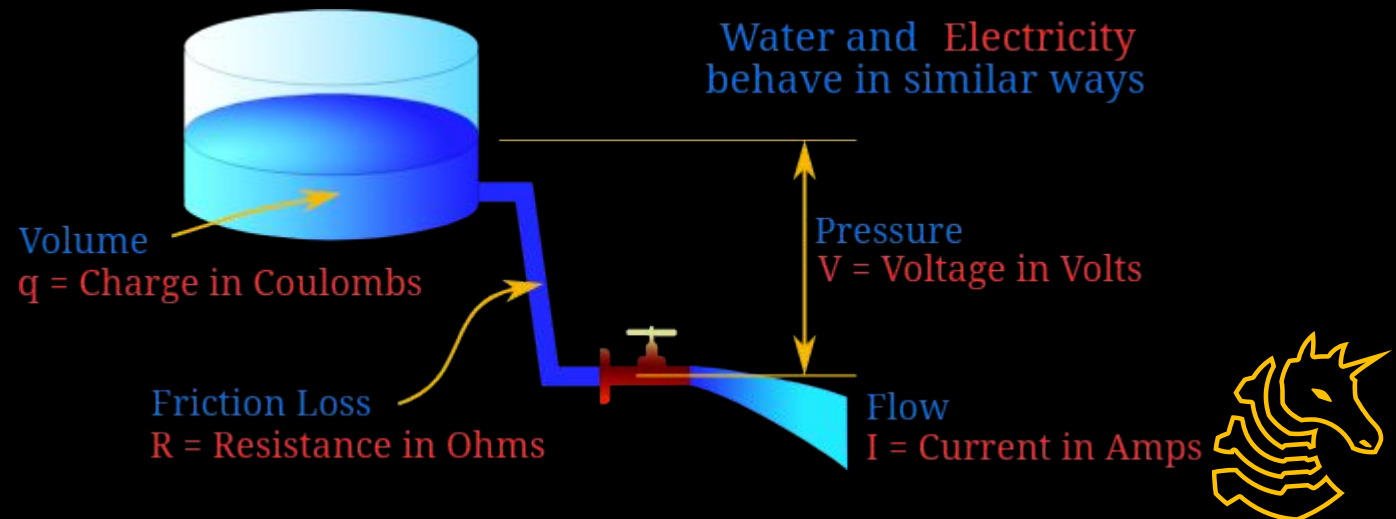
$$V = I R$$



$$I = \frac{V}{R}$$

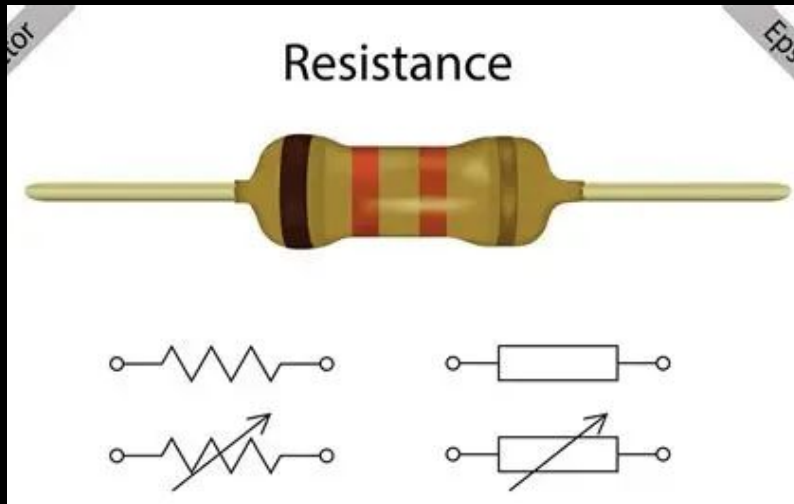


$$R = \frac{V}{I}$$

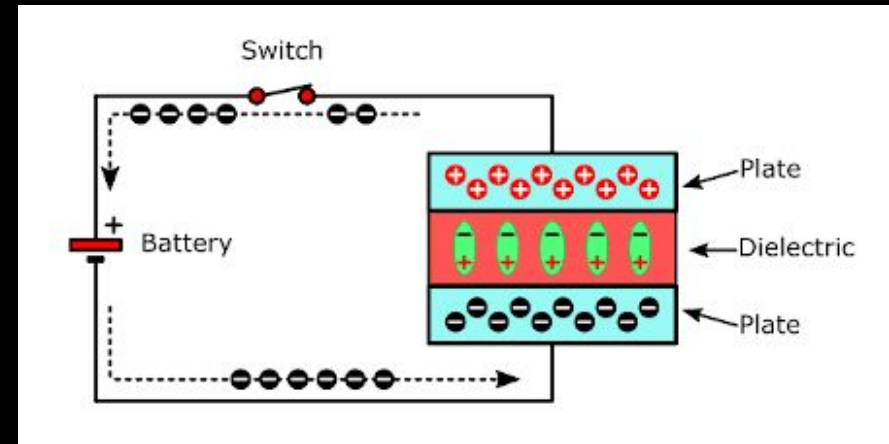


Circuit Elements

- **Resistor** - introduces resistance to current in a circuit
 - Serves to control current flow and therefore the functionality of a circuit
 - Think: a narrow pipe that water flows through

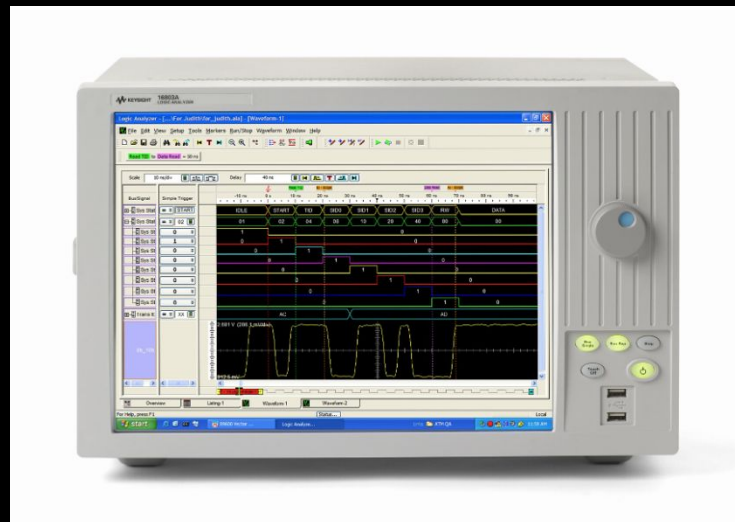


- **Capacitor** - stores electrical energy in the form of an electric field
 - Parallel plates accumulate opposite charges, forming electric field
 - Think: a water tank/buffer

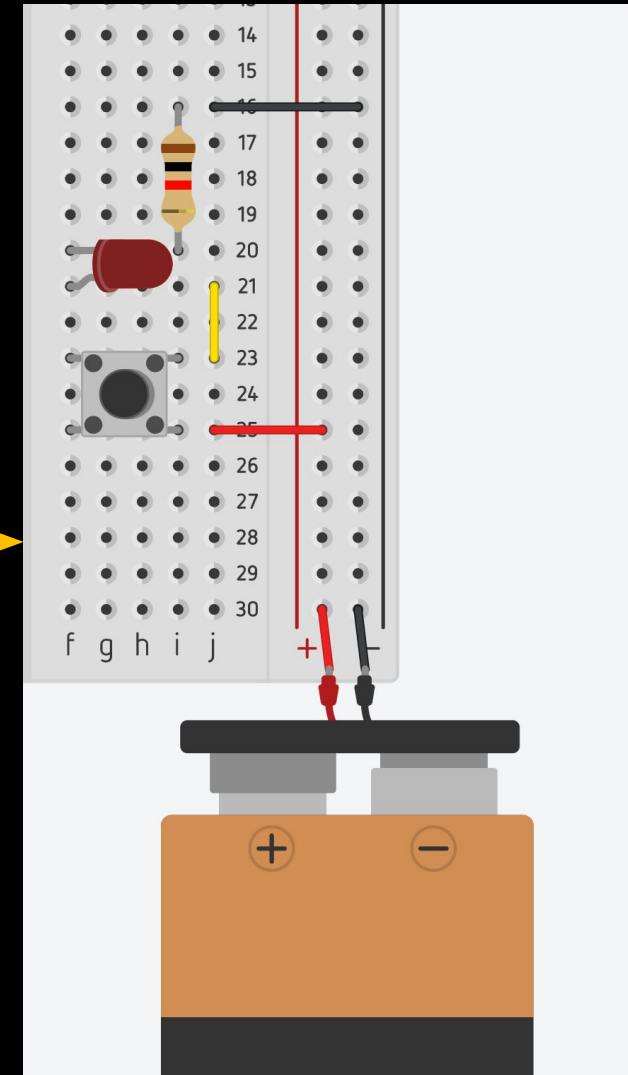
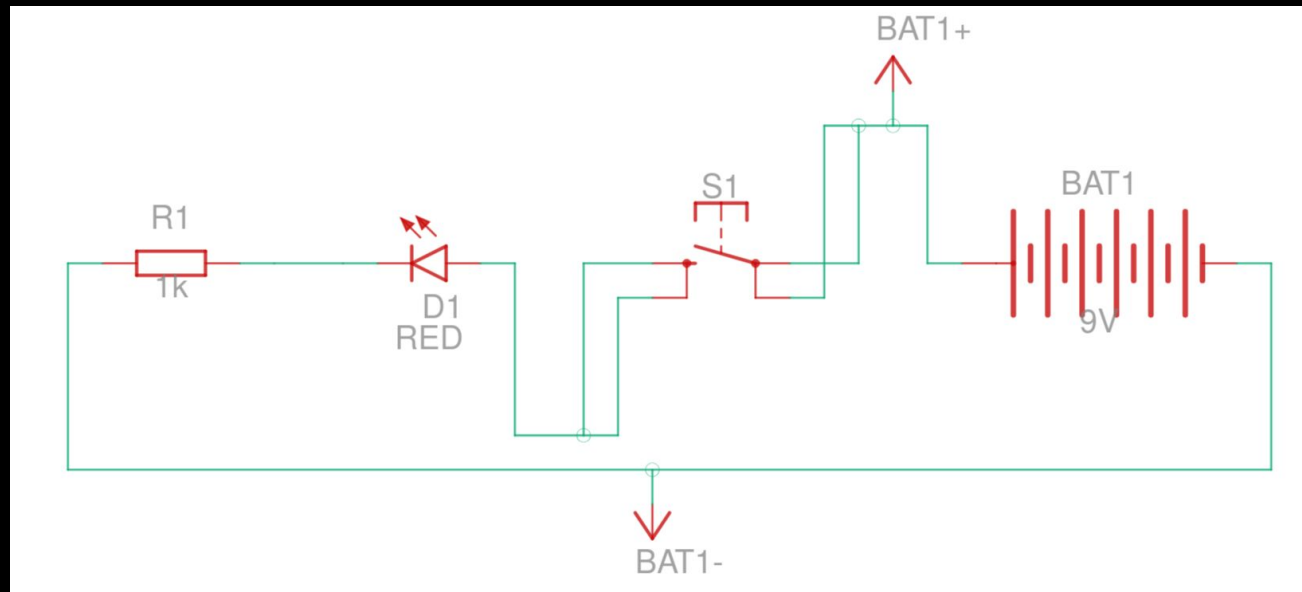


Measurement & Analysis Tools

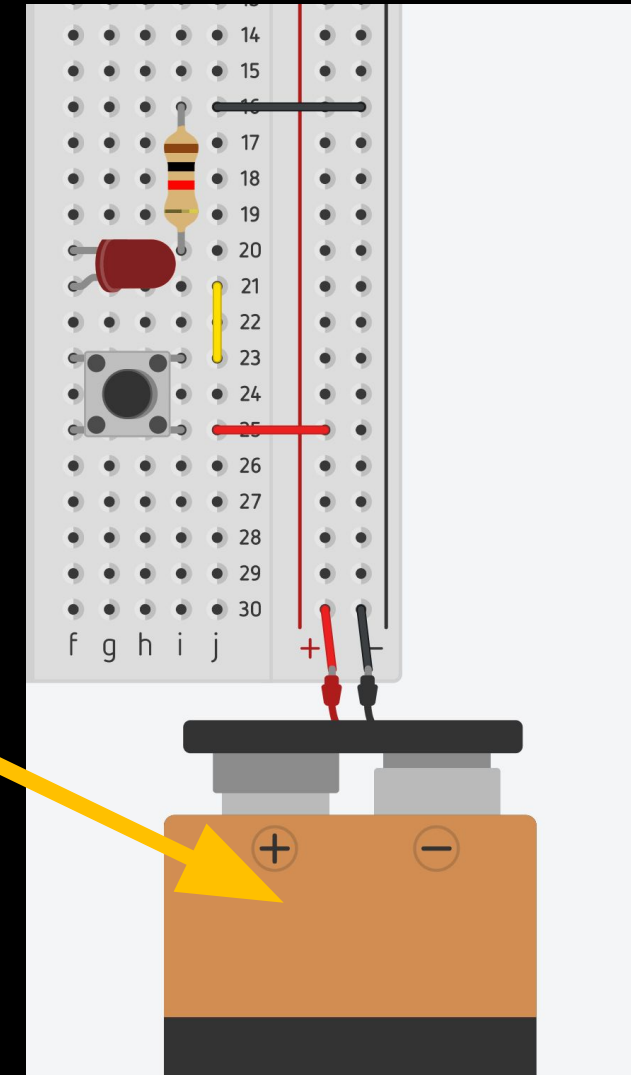
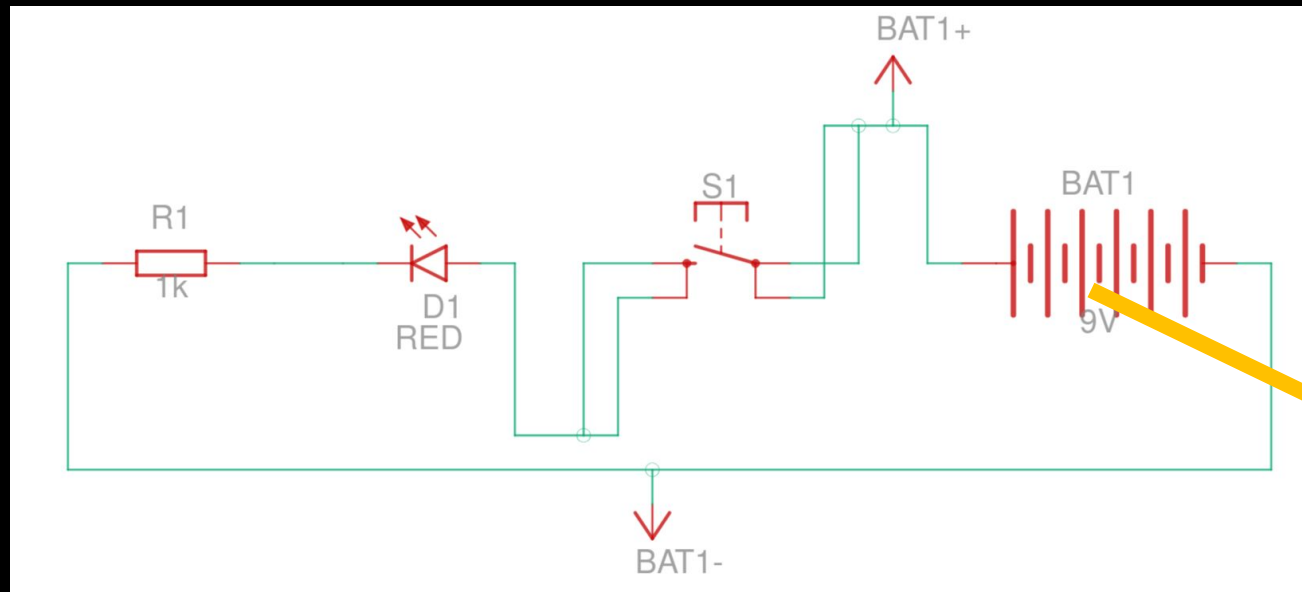
- **Oscilloscope** - instrument used to measure analog quantities from a circuit as chosen/defined by the user
- Probes can be connected to any parts of a circuit to measure voltage, current, etc. between 2 points
- **Logic analyzer** - instrument used to measure digital signals for circuits (logic 1 and 0)
- Voltage above certain threshold considered “high” and below certain threshold considered “low”
- **ChipWhisperer** - board used to read power traces and perform side channel analysis and fault injection on other devices



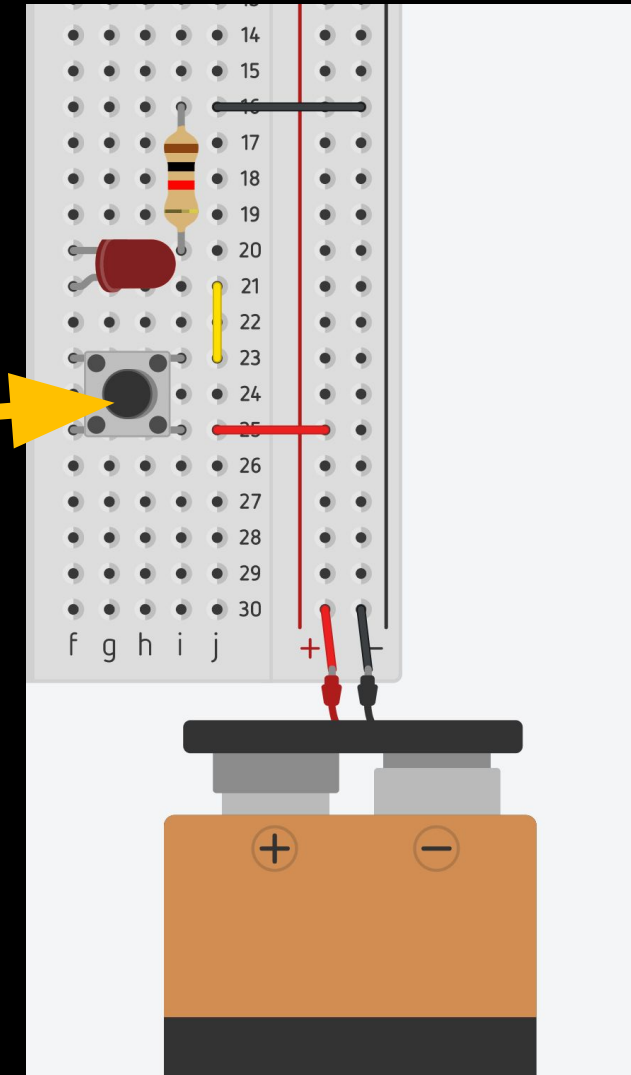
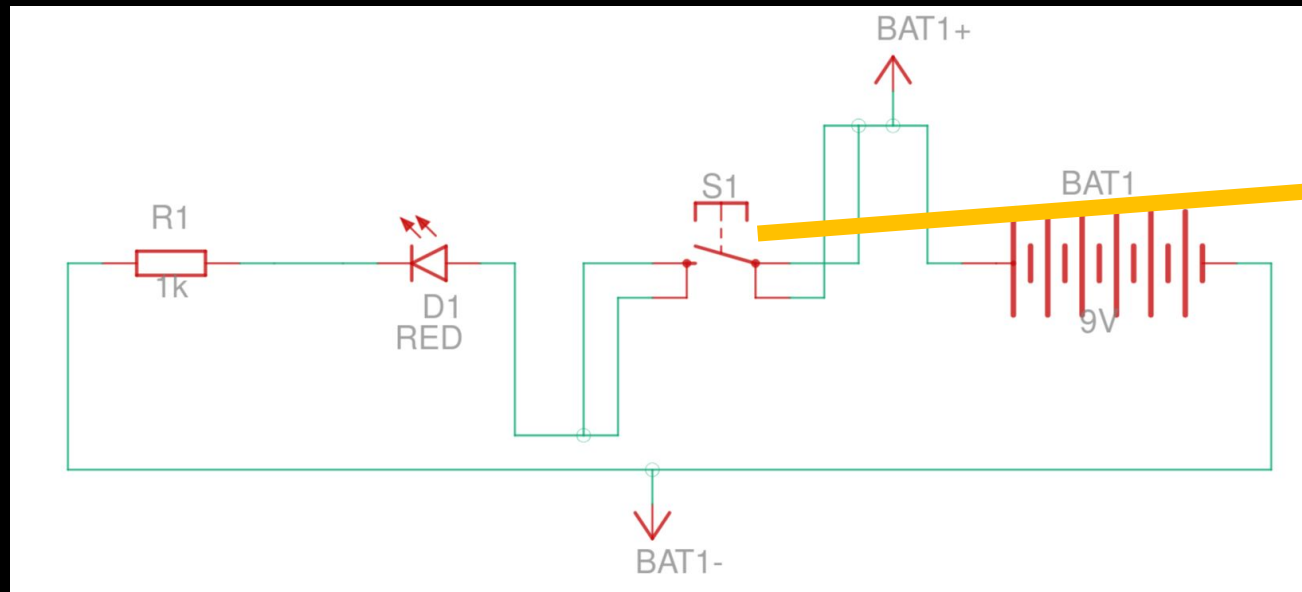
Reading a circuit diagram



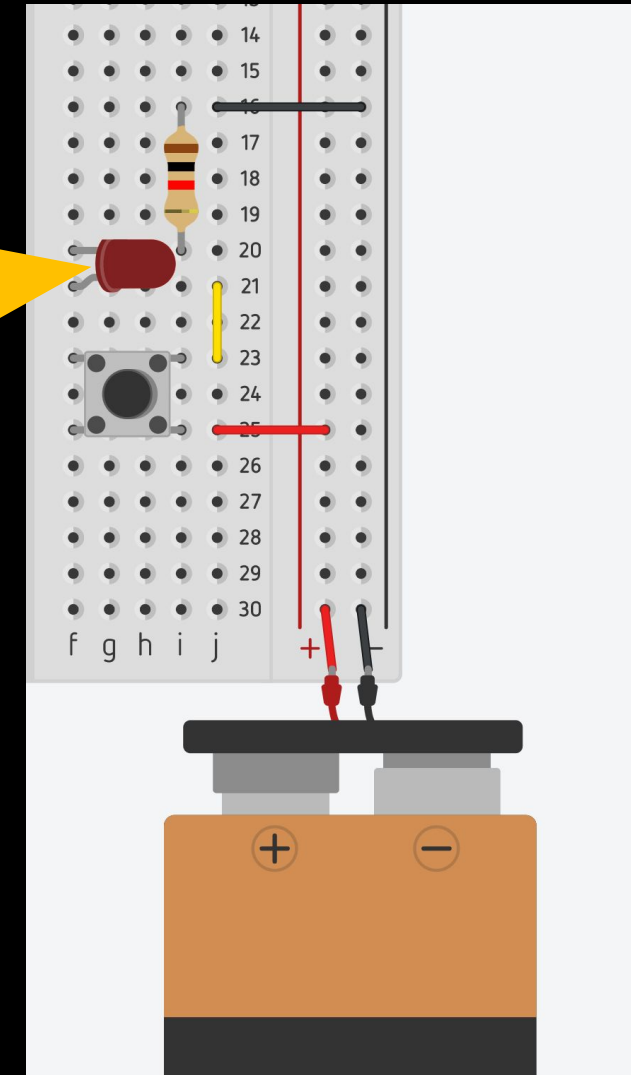
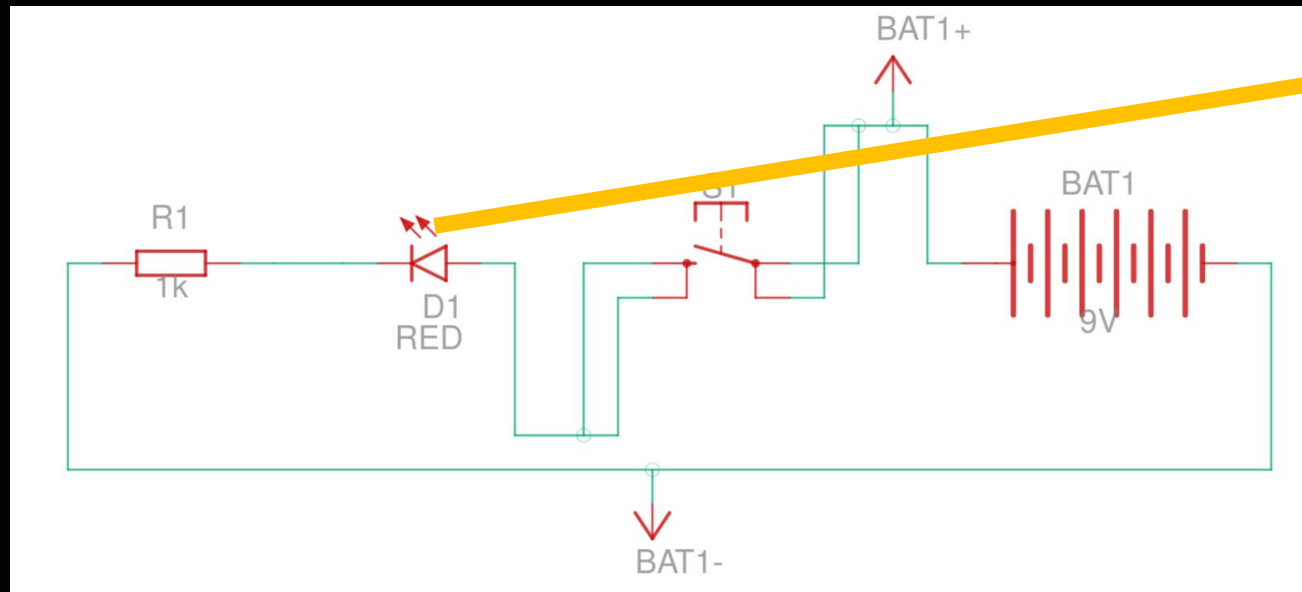
Reading a circuit diagram



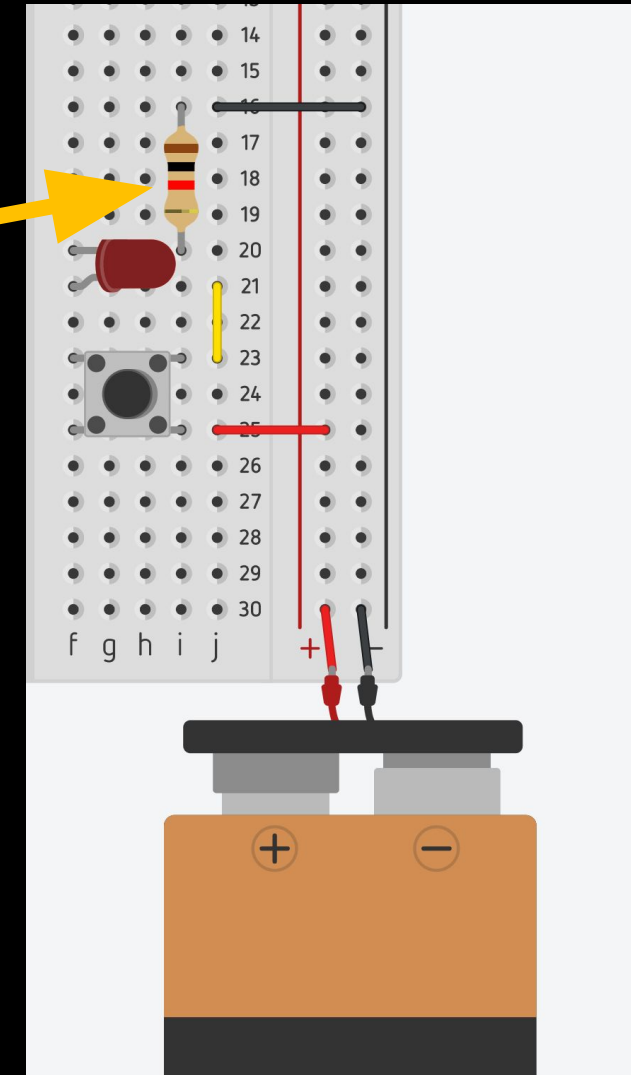
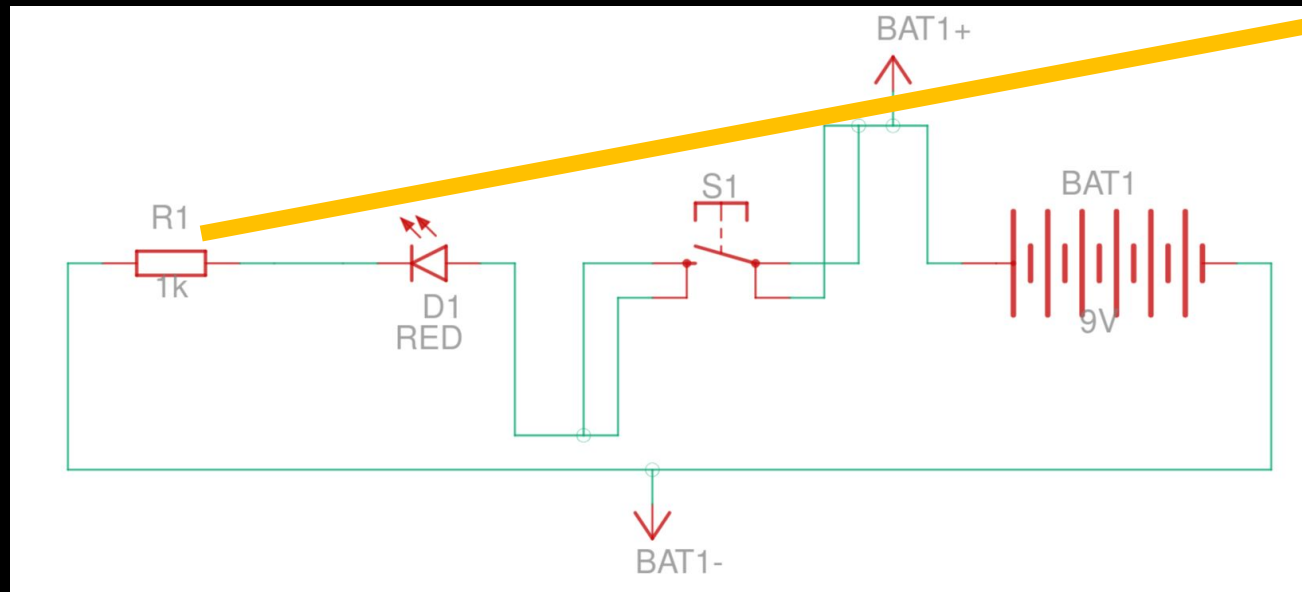
Reading a circuit diagram



Reading a circuit diagram



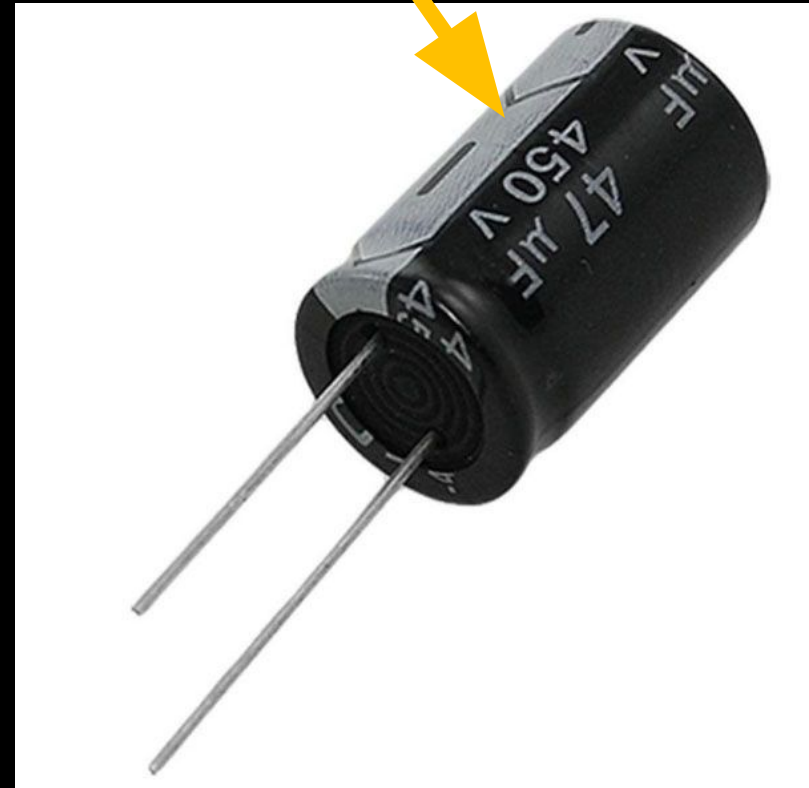
Reading a circuit diagram



Warning: polarized capacitors

- The **long** end goes to +
- **Short** end goes to -
- Backwards won't work and may damage the capacitor

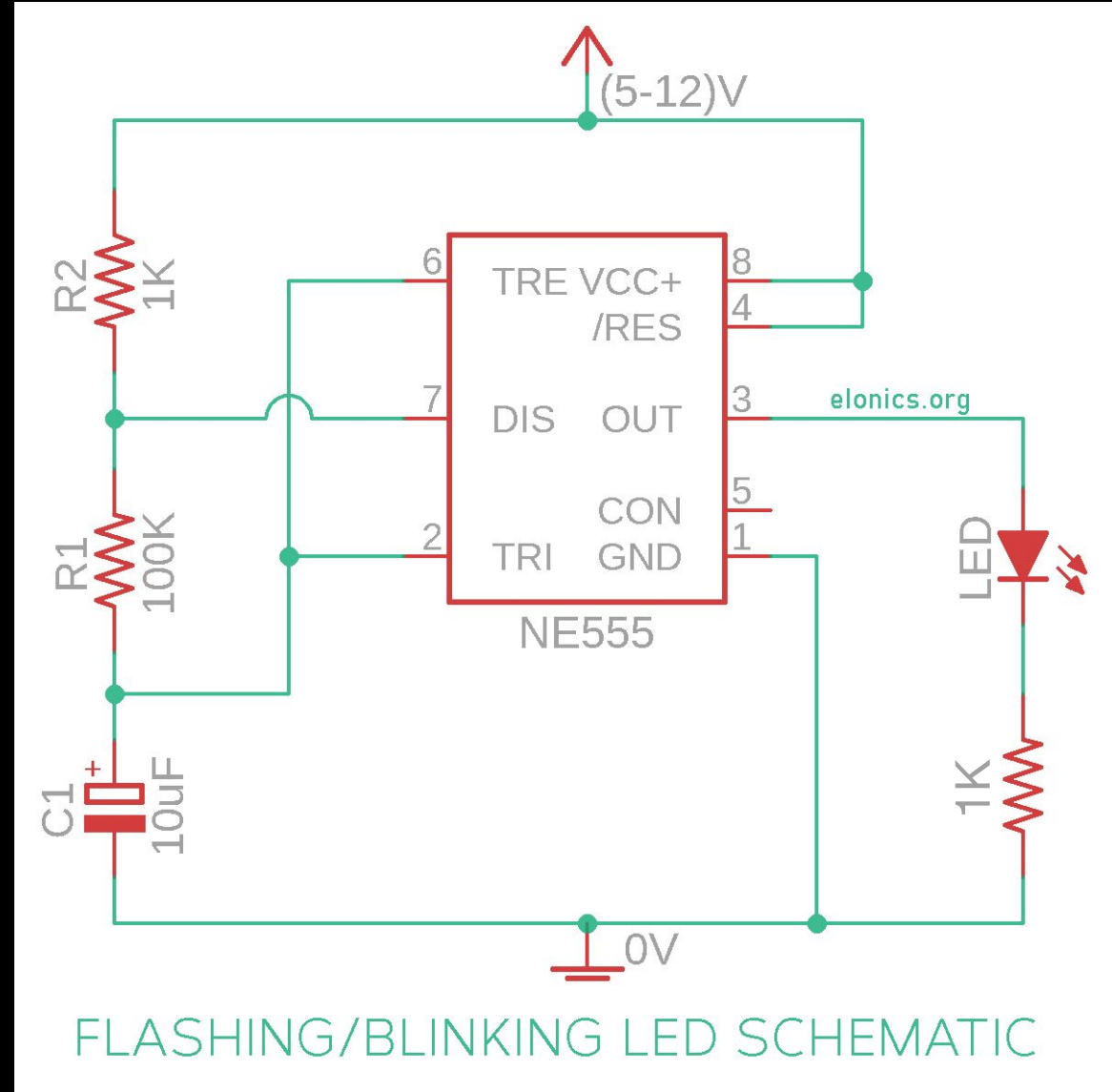
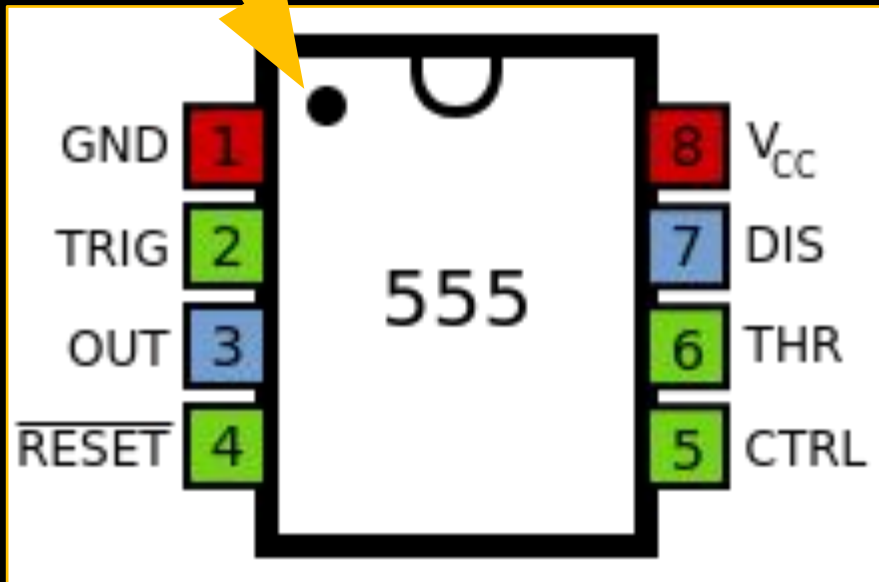
Polarity band



Today's Exercise

- Astable 555 Blinker circuit
 - Blinky light

orientation dot



Today's Exercise

- Astable 555 Blinker circuit
 - Blinky light
- 555 IC provides a PWM (pulse-width modulation) signal similar to a clk signal
- The LED will turn on when the signal from the 555 is “high”

Datasheets:

- [TLC555CP T106AN datasheet](#)
- [19AHN6M NE555P datasheet](#)



Resources at UIUC

- SIGPwny Embedded Team bench (ACM back room)
 - Ask before taking anything, otherwise, it's free to use for SIGPwny/ACM
- IoT Makerspace (Siebel CS 1109, right across from ACM room)
- OpenLab (ECEB 2024)
- Electronic Services Shop (1st floor ECEB)
 - FREE!!!
 - Self-service, better hours than supply center
- ECE Supply Center (1st floor ECEB)
 - If E-Shop doesn't have what you need, you can buy from here





Next Meetings

2026-10-12 • Next Monday

- Microcontroller programming lab (once our bulk order of Launchpad boards arrives!)



Meeting content can be found at
sigpwny.com/meetings.

